



Mubeen Afzal

Date of birth: 02/12/2001 | **Nationality:** Pakistani | **Gender:** Male | **Phone number:** (+49) 15214055115 (Mobile) | **Email address:** mubeen.afzal2001@gmail.com | **Website:** mubeen.live | **LinkedIn:** [mubeen-afzal-ai](https://www.linkedin.com/in/mubeen-afzal-ai) | **Address:** Saarbrücken, 66125, Saarland, Germany (Home)

● ABOUT ME

AI Engineer with hands-on experience in building and deploying scalable solutions in computer vision, natural language processing, and MLOps. Passionate about continuous learning and innovation, with a strong interest in combining AI and cybersecurity to advance threat detection, vulnerability analysis, and secure automation.

● EDUCATION AND TRAINING

01/10/2025 – CURRENT Saarbrücken, Germany
MASTERS IN CYBERSECURITY Saarland University

Website <https://www.uni-saarland.de/en/home.html>

07/2019 – 01/2024 Islamabad, Pakistan
BACHELORS IN ARTIFICIAL INTELLIGENCE National University of Computer and Emerging Sciences

Website <https://www.nu.edu.pk/>

● SKILLS

Computer Languages

Python | Javascript | C++ | GO

Database System

vector databases | SQL | NoSQL

Artificial Intelligence

Machine Learning | Deep Learning | Generative AI | Agentic AI | NLP | Computer Vision | Data Wrangling | Data Aanalysis | Data Scraping

Others

Linux | Azure | AWS | Dockers | Git | Microsoft Office

Soft Skills

Communication Skills | Leadership | Time Management

● WORK EXPERIENCE

🏢 **IKONIC – ISLAMABAD, PAKISTAN**
AI ENGINEER – 06/2025 – 08/2025

- Implemented multi-agent systems with LangChain, LangGraph, and Model Context Protocol (MCP), enabling agent-to-agent and agent-to-tool communication.
- Developed automation workflows using no-code tools to streamline repetitive tasks and improve efficiency.
- Engineered scalable architectures for agentic systems, focusing on interoperability and modular integration.

🏢 **IMARAT GROUP OF COMPANIES – ISLAMABAD, PAKISTAN**

AI ENGINEER – 05/2024 – 11/2024

- Built an AI-powered architectural design tool leveraging diffusion models and ControlNet to automate floor plan generation and house design rendering.
- Implemented style-based customization features to generate diverse architectural outputs.
- Improved real-time interactivity and optimized system performance for scalable AI deployments.

🏢 QLU.AI – ISLAMABAD, PAKISTAN

DEEP LEARNING ENGINEER – 01/2024 – 04/2024

- Designed a Retrieval-Augmented Generation (RAG) chat system using vector databases for efficient data retrieval and transformer models for high-quality responses.
- Refactored AI service codebase to improve scalability and simplify integration with other systems.
- Implemented model monitoring to track performance and reliability in production.

🏢 DATAINSIGHT LAB – ISLAMABAD, PAKISTAN

COMPUTER VISION RESEARCH INTERN – 06/2023 – 08/2023

- Built multilingual handwritten text recognition pipelines for medical records, improving model accuracy.
- Designed efficient data processing workflows to handle large datasets for training and validation.
- Optimized model performance through preprocessing, transformation, and validation strategies.

● PROJECTS

Machine Learning-Enhanced Threat Intelligence Platform

Collaborated with CydeaTech for FYP thesis to develop a cutting-edge threat intelligence system that enhances enterprise cybersecurity. The platform integrates **machine learning-based anomaly detection** (using **autoencoders** and **Isolation Forest** algorithms) with **rule-based techniques** to deliver robust threat detection and response. Built with **Python** for backend processing, **JavaScript** for interactive visualizations, and the **ELK Stack** for log management and analysis, the system was designed and validated using real-world **SIEM logs**. Through advanced **feature engineering**, it achieved superior accuracy and reduced detection time for security incidents, strengthening enterprise threat intelligence capabilities.

AI-Powered Floor Plan and House Rendering Platform

Engineered an innovative system to empower end users and professional architects in designing and rendering floor plans and houses through intuitive selectable options and prompts. Leveraged **Graph Neural Networks (GNNs)** for intelligent floor layout generation, **custom-trained diffusion models** for high-fidelity house rendering, and **ControlNet** to precisely control house structure and design elements. Developed using **Python** for backend model training and processing, **JavaScript** for interactive front-end interfaces, and **Django** as the framework for seamless system integration. Created a proprietary dataset to train models, ensuring high-quality, contextually accurate outputs. Achieved rendering times of up to 10 seconds on consumer GPUs, enabling efficient and accessible architectural design for diverse users.

Intelligent Executive Interview Platform

Developed an advanced interview system to streamline executive-level recruitment processes. Built with **Python**, **OpenAI's ChatGPT**, **Whisper** for speech-to-text transcription, **Twilio** for real-time communication, and **FastAPI** for efficient API development. The platform dynamically supports interviewers by generating in-depth follow-up questions using natural language processing (NLP) and managing predefined interview agendas. Key features include real-time transcription of candidate responses, intelligent question recommendations, and seamless agenda transitions, ensuring structured and thorough evaluations. This solution enhances interviewer efficiency and improves candidate assessment quality for high-level executive roles.

Retrieval-Augmented Generation (RAG) System for Email Query Processing

Developed a Retrieval-Augmented Generation (RAG) system to enable intelligent querying and response generation from user email data. Built using **Python** and **FastAPI** for efficient API development, the system integrated a **custom-trained Large Language Model (LLM)** for natural language understanding and response generation. Utilized **Pinecone** vector database for semantic search and **Redis** for caching to optimize query performance. The platform allowed users to extract insights and answers from their mailbox data efficiently. Development was halted due to security concerns regarding sensitive email data handling, ensuring compliance with privacy standards.

Multi-Agent AI-Powered Trip Planning

Developed an advanced multi-agent AI system leveraging **Agentic AI** and **Model Context Protocol (MCP)** to enable seamless communication between agents and external tools for enhanced trip planning. Built with **Python** and **FastAPI** for secure API communication, the system utilized **LangChain** and **LangGraph** for agent orchestration and workflow management, **PostgreSQL** for robust conversation storage, and **Redis** for efficient caching. Integrated real-time data from flight, hotel, and **Google Maps** APIs, incorporating reviews and ratings to provide personalized travel recommendations.

● **CERTIFICATIONS**

Google
Google Cybersecurity

Microsoft
Azure Fundamentals

DeepLearning.ai
Generative AI with LLMs

DeepLearning.ai
Introduction to Generative AI

● **PUBLICATIONS**

• **AI-Enhanced Architectural Design: From Interactive Floor Plan Generation to Style Rendering Using Diffusion Models and ControlNet**

A comprehensive AI framework for automating floor plan creation with real-time user interaction and style-based rendering, leveraging diffusion models to transform architectural workflows. (In-Progress)

● **LANGUAGE SKILLS**

Mother tongue(s): **URDU**

Other language(s): **ENGLISH**